

SIMATS ENGINEERING

ASSESSMENT TOOL 3: Application-Based Questions

Course Name: Cloud Computing and Big Data Analytics

Course Code: CSA15

Course Outcome Covered:

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Question 1

A company experiences unpredictable traffic spikes during seasonal sales.

Tasks:

- Explain how cloud auto-scaling and load balancing can handle this demand efficiently.
- How does this approach reduce downtime and improve user experience?
- What role do monitoring tools play in this setup?
- Relate your answer to a real-world e-commerce scenario.

Answer

Introduction

During seasonal sales such as Black Friday or festive discounts, e-commerce websites experience sudden increases in customer traffic. Cloud computing provides auto-scaling and load balancing to manage these traffic spikes efficiently without affecting application performance.

Auto-Scaling

Auto-scaling automatically adjusts the number of virtual machines or containers based on demand.

- Adds new instances during high traffic.
- Removes extra instances when traffic decreases.
- Optimizes resource utilization.
- Reduces infrastructure costs.

Load Balancing

A load balancer distributes incoming requests across multiple servers.

- Prevents server overload.
- Improves application response time.
- Ensures continuous availability.
- Eliminates single points of failure.

Benefits

- Reduced downtime.
- Faster website loading.
- Better customer experience.
- Higher sales during peak demand.

Role of Monitoring Tools

Monitoring tools continuously observe:

- CPU utilization.
- Memory usage.
- Network traffic.
- Server health.
- Response time.

Alerts are generated automatically when predefined thresholds are exceeded, allowing auto-scaling to respond quickly.

Real-World Example

During Amazon Prime Day, millions of users access the platform simultaneously. Auto-scaling launches additional servers, load balancers distribute traffic evenly, and monitoring tools detect performance issues in real time, ensuring uninterrupted shopping.

Conclusion

Auto-scaling, load balancing, and monitoring together provide a highly available and scalable cloud environment that delivers excellent performance during traffic spikes.

Question 2

An organization wants to store large volumes of backup data securely in the cloud.

Tasks:

- Discuss how cloud storage solutions ensure durability and data redundancy.
- How can encryption and access control protect sensitive backup data?
- Explain the cost advantages compared to traditional storage systems.
- Provide an example of a disaster recovery use case.

Answer

Introduction

Cloud storage enables organizations to store backups securely while ensuring high durability, redundancy, and cost efficiency.

Durability and Data Redundancy

Cloud providers maintain multiple copies of backup data across different storage devices and geographical regions.

- Automatic replication.
- Data integrity verification.
- Fault tolerance.
- High availability.

Encryption

- Encrypt data during transmission using TLS/SSL.
- Encrypt stored data using AES-256.
- Protect encryption keys using Key Management Services (KMS).

Access Control

- Role-Based Access Control (RBAC).
- Multi-Factor Authentication (MFA).
- Identity and Access Management (IAM).
- Audit logging.

Cost Advantages

- No investment in physical storage hardware.
- Pay only for storage used.
- Automatic scalability.
- Lower maintenance and operational costs.
- Reduced power and cooling expenses.

Disaster Recovery Example

A company's primary datacenter is damaged due to flooding. Cloud backups stored in another region are restored immediately, allowing business operations to continue with minimal downtime and data loss.

Conclusion

Cloud storage provides secure, durable, and economical backup solutions while ensuring rapid disaster recovery and business continuity.

Question 3

A mobile app company plans to deploy its application globally.

Tasks:

- Explain how Content Delivery Networks (CDNs) improve performance and reduce latency.
- How do edge servers enhance user experience across regions?
- Discuss how cloud providers support global availability.
- Relate this to a real-time streaming or gaming application.

Answer

Introduction

Global applications require fast content delivery to users regardless of their location. Cloud providers use Content Delivery Networks (CDNs) and edge servers to improve application performance.

Content Delivery Networks (CDNs)

A CDN stores copies of content on servers distributed worldwide.

- Reduces latency.
- Decreases bandwidth usage.
- Improves loading speed.

- Reduces server workload.

Edge Servers

Edge servers are located close to end users.

Benefits include:

- Faster response time.
- Reduced network congestion.
- Improved user experience.
- Lower buffering for multimedia content.

Global Availability

Cloud providers maintain data centers across multiple regions.

Features include:

- Multi-region deployment.
- Global load balancing.
- Automatic failover.
- Geographic redundancy.

Real-World Example

A live sports streaming application delivers video through CDN edge servers. Users in Asia, Europe, and North America receive content from nearby servers, reducing buffering and ensuring smooth playback.

Conclusion

CDNs and edge computing significantly improve application speed, reduce latency, and provide reliable global access for users.

Question 4

A financial institution requires strict compliance and data security in cloud usage.

Tasks:

- Explain how cloud providers support regulatory compliance and auditing.
- What role do logging, monitoring, and encryption play in security?
- How can the company ensure data sovereignty requirements are met?
- Give an example involving sensitive financial transactions.

Answer

Introduction

Financial institutions handle highly sensitive customer information and must comply with strict regulations related to data security and privacy.

Regulatory Compliance

Cloud providers support standards such as:

- ISO 27001.
- PCI-DSS.
- SOC 2.
- GDPR.
- HIPAA (where applicable).

These certifications demonstrate adherence to security and privacy best practices.

Logging and Monitoring

Logging records:

- User login activities.

- File access.
- Configuration changes.
- Security events.

Monitoring detects:

- Unauthorized access.
- Suspicious activities.
- System failures.
- Performance issues.

Encryption

- Encrypt data at rest using AES-256.
- Encrypt data in transit using TLS/SSL.
- Use secure key management services.

Data Sovereignty

Organizations can:

- Store data within approved geographic regions.
- Choose cloud regions based on legal requirements.
- Restrict cross-border data movement.
- Maintain compliance with national regulations.

Example

An online banking application encrypts every financial transaction, records all activities in audit logs, continuously monitors suspicious behavior, and stores customer data within the country's approved cloud region to comply with financial regulations.

Conclusion

Cloud providers help financial institutions maintain regulatory compliance through encryption, auditing, monitoring, and regional data storage while protecting sensitive customer information.

Question 5

A development team wants to adopt containerization for application deployment.

Tasks:

- Explain how containers differ from virtual machines in terms of performance and resource usage.
- How does orchestration (like Kubernetes) simplify deployment and scaling?
- Discuss portability benefits across different cloud environments.
- Provide an example of a microservices-based application.

Answer

Introduction

Containerization packages applications and their dependencies into lightweight, portable units that can run consistently across different environments.

Containers vs. Virtual Machines

Feature	Containers	Virtual Machines
Operating System	Share host OS	Separate guest OS
Startup Time	Seconds	Minutes
Resource Usage	Low	High
Performance	Near-native	Moderate

Portability	High	Moderate
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Scalability	Excellent	Good
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Kubernetes Orchestration

Kubernetes automates:

- Container deployment.
- Load balancing.
- Auto-scaling.
- Service discovery.
- Self-healing.
- Rolling updates.

Portability Benefits

Containers can run consistently across:

- Private clouds.
- Public clouds.
- Hybrid clouds.
- Multi-cloud environments.

This eliminates environment compatibility issues and simplifies application migration.

Microservices Example

An online shopping application can be divided into independent microservices such as:

- User Authentication Service.
- Product Catalog Service.

- Shopping Cart Service.
- Payment Service.
- Order Management Service.
- Notification Service.

Each microservice runs in its own container and is managed by Kubernetes, enabling independent deployment, scaling, and maintenance.

Conclusion

Containerization and Kubernetes improve application performance, reduce resource consumption, simplify deployment, and provide excellent portability across cloud platforms. They are ideal for modern microservices-based cloud applications.